

THE GENOME

A GENOME IS THE FULL SET OF GENETIC INSTRUCTIONS FOR A LIVING THING. THE HUMAN GENOME CONSISTS OF AN ESTIMATED 20,000 GENES FOR MAKING PROTEINS, CARRIED ON THE DOUBLE SET OF 46 CHROMOSOMES IN MOST BODY CELLS.



KARYOTYPE

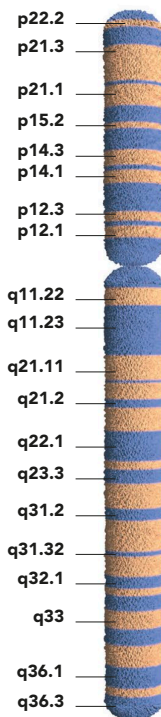
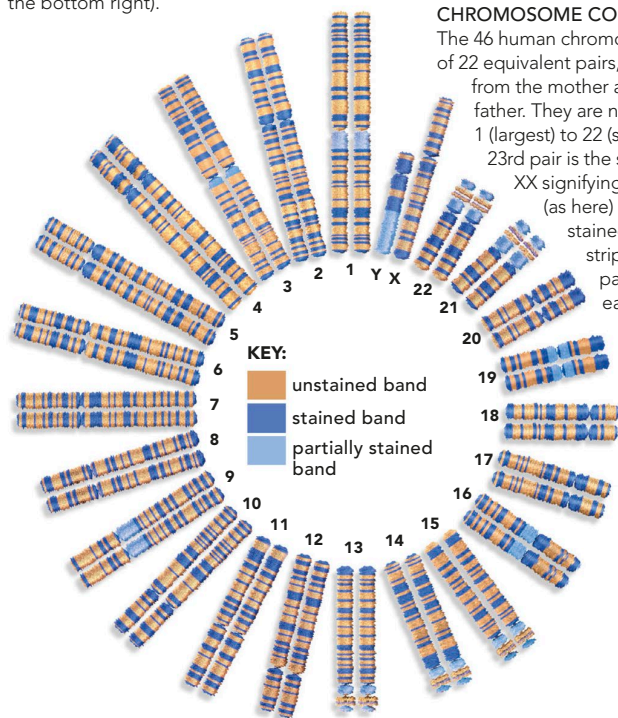
A karyotype is a photograph of all chromosomes from a cell arranged in a standard order. This example is from a female (note the two, equal-sized X chromosomes at the bottom right).

CHROMOSOMES AND DNA

The Human Genome Project, a multinational effort to map the human genome, was completed in 2003. It led to the identification of more than 20,000 individual human genes within the 46 chromosomes that collectively include about 3.2 billion base pairs. Although much of the DNA that makes up the chromosomes does not code for proteins, known as non-coding and “junk” DNA, it may still regulate gene function. Junk DNA is different from non-coding DNA in that its structure does not resemble that of genes.

CHROMOSOME COMPLEMENT

The 46 human chromosomes consist of 22 equivalent pairs, one of each pair from the mother and one from the father. They are numbered from 1 (largest) to 22 (smallest). The 23rd pair is the sex chromosomes, XX signifying female and XY (as here) male. When stained with chemicals, stripes called banding patterns show up on each chromosome.



CHROMOSOME SEVEN

This chromosome contains more than 5 per cent of the genome's total DNA, with about 159 million pairs of bases. Almost 60 million are in the short arm, 7p, with the rest in the longer arm, 7q.